

MATE ROV Technical Documentation

Tuesday, March 11, 2025 12:05 PM

*Need tech team's input

*Pages

Outline

- 1: Cover Page with member names
- 2: Abstract --> Clair
- 3: Table of Contents
- 4 – 5: Teamwork --> Yolanda
 - Company Organization
 - Company structure
 - Project Management
 - What communication tools used to collaborate on the ROV?
 - Asana, Fusion, etc.
 - Please explain briefly what each are used for
 - Provide screenshots if
 - Project Timeline
 - From training of new members --> sorting into different technical teams --> design --> fabrication --> testing
 - Provide the timeline image
- Design Rationale --> Clair, Ian, Nick, Sakiba, Charlotte
 - 6: Vehicle Overview
 - ROV Innovations --> What's new? --> back up the ideas using real-life scenarios
 - Overview of the ROV's features --> no need to get into detail
 - 7 – 8: Structure & Support
 - Frame
 - Chamber
 - PCB Mounting?
 - Any other structural support used inside the chamber?
 - Support
 - Stand at the bottom of the ROV for stability at ocean floor
 - Can we do a simple simulation of the stand or something to show that it's useful (not just for task)?
 - 9:
 - Propulsion
 - Thruster configuration
 - Support 6 degrees of freedom
 - Buoyancy
 - Any center of mass/center of buoyancy calculations?
 - Are foam blocks used for buoyancy? Where and why that location?
 - Any other ways the ROV is optimized for buoyancy?
 - Electronics
 - 10:
 - Tether
 - Tether Cross-Section Diagram
 - Electronic Chamber --> Include power conversion details for components below
 - What specific PCBs are used, and for what?
 - ESC, Microcontrollers, Raspberry Pi for thrusters
 - Cameras, sensors connected
 - Gripper modules connected
 - How is the chamber kept organized? What kind of attachments to hold down/bundle the wires inside?
 - 11 – 12:
 - PCB Design
 - We designed our own PCBs, why?
 - Specific benefits of designing it ourselves vs buying from the market?
 - Please provide a brief explanation of the PCB specs
 - Power Consumption
 - Calculations of how much power is required by the ROV for all of its features --> Refer to Purdue Tech Doc p. 11/25
 - Do you guys want to include this?
 - Control Station
 - Diagram
 - Power Supply
 - Overview and description
 - Include a picture

- 13 – 14: Software
 - Piloting UI
 - What can the pilots control?
 - How do the pilots control the ROV?
 - What features are included?
 - Camera Views
 - Camera positions
 - Specifically explain how the camera views on the interface will help remote operation of the ROV
 - Camera calibration
 - Computer Vision
 - Detail what the software analyzes
 - What is the input, what is the output
 - How is this useful
- 15: Grippers (Ian)
 - Overview: detail the modular aspect of the gripper design on the current iteration of the ROV
 - Emphasize that this is an improvement from last year to fulfill some pain points
 - Then, explain the mechanism of each gripper
- 16: Safety --> Copy and adapt content from last year's tech doc
 - Philosophy
 - Training
 - Protocols and Standards
 - Features
- 17: Accounting --> Made by Donald
- 18: Acknowledgements & References --> Clair
 - Sponsorship
 - Acknowledgements for other partners --> if necessary

Pick your parts (make sure your parts total up to at least 2 pages):

- Teamwork (p.4-5): Yolanda
- Vehicle Overview (p.6): Clair
- Structure & Support (p.7-8): Nick
- Propulsion, Buoyancy (p.9): Ian
- Tether, Electronic Chamber (p.10): Clair
- PCB Design, Power Consumption, Control Station, Power Supply (p.11-12): Sakiba
- Software (p.13-14): Charlotte
- Grippers (p.15): Ian
- Safety (p.16): Ian